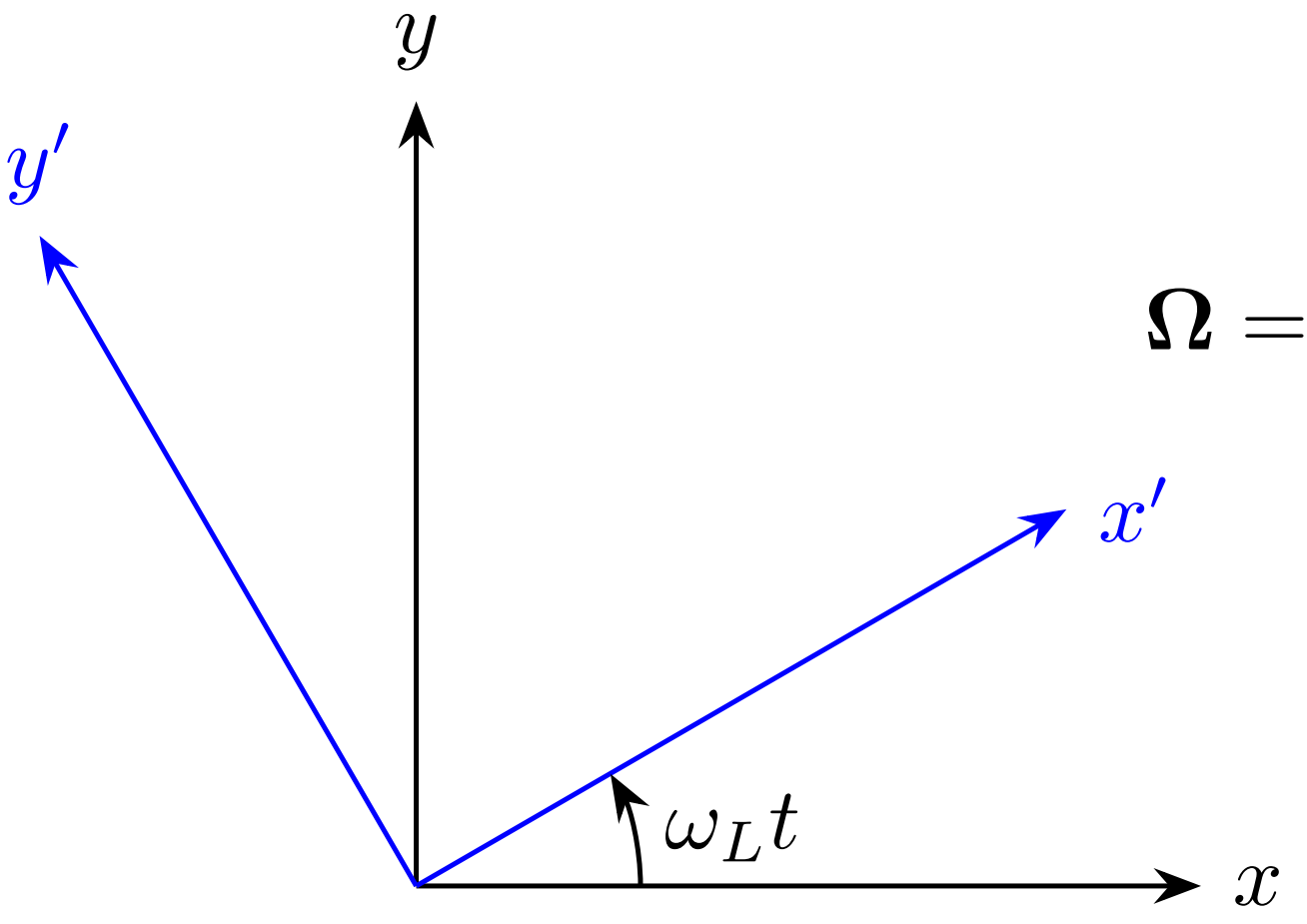

$$\mathbf{\Omega} = \omega_L \mathbf{k}$$

$$\mathbf{v} = \mathbf{v}' + \mathbf{\Omega} \times \mathbf{r}$$